

MUHAMMAD ANAS

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building LLM-powered applications, multi-agent pipelines, and production ML systems in Python. Built projects spanning multimodal vector search, MCP-based BI assistants, and agentic shopping advisors using LangGraph, Gemini API, and FastMCP. Comfortable working with REST APIs, Git-based collaborative workflows, and prompt engineering for real-world AI use cases.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++

ML / AI: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, CLIP

LLM / Agents: LangGraph, LangChain, Google Gemini API, Ollama (Llama 3.2), FastMCP, Prompt Engineering

Backend & DBs: FastAPI, Flask, PostgreSQL, MongoDB Atlas, Weaviate (Vector DB), Redis, REST APIs

Tools: Git, Docker, Playwright, BeautifulSoup, Jupyter Notebook, Railway, Supabase

EXPERIENCE

Machine Learning Trainee — Elevvo Pathways (Remote, Egypt)

Aug 2025 – Sep 2025

- Trained classification and regression models on 50K–200K-record datasets using TensorFlow and Scikit-learn; managed the full pipeline from data cleaning, preprocessing, and feature engineering through cross-validation and evaluation.
- Built CNN-based image recognition systems for a multi-class visual task; applied custom data augmentation and achieved 88% validation accuracy.

PROJECTS

Findify — Multimodal AI Product Discovery Platform ↗

Nov 2024 – Jul 2026

Flutter • FastAPI • MongoDB Atlas • Weaviate • CLIP • Redis • Playwright

- Engineered end-to-end multimodal search (text + image) using CLIP embeddings and Weaviate named vectors, enabling product discovery by photo or natural-language description — sub-100ms retrieval across 7,000+ products validated at 250+ concurrent RPS.
- Built parallel Playwright scraping pipeline (Amazon, eBay, Alibaba) with ThreadPoolExecutor; reduced per-batch enrichment time ~60%, sourcing 7,000+ listings; Upstash Redis caching layer cut average retrieval latency 85% (35ms → <5ms).
- Resolved 4 production-blocking failures (Weaviate ingestion mismatch, ISP port-blocking, Railway RAM overflow, env var mismatches), bringing system from 0% to full search functionality.

MCP-Based SQL Data Analyst Assistant ↗

May 2026

FastMCP • PostgreSQL • Gemini 2.5 Flash • Next.js • YAML Semantic Layer

- Built a conversational NLP-driven BI assistant allowing non-technical users to query a 2.5M-row retail warehouse in plain English, eliminating dependency on analysts for routine reporting.
- Implemented an MCP server with 11 capabilities as a read-only LLM gateway; achieved 100% adversarial block rate across all SQL injection and PII-extraction tests.
- Designed a YAML semantic layer mapping business terms to SQL metric formulas and safe join paths, eliminating schema ambiguity for the LLM.

AI-Powered Intelligent Shopping Advisor ↗

2025

Python • LangGraph • Gemini API • Ollama (Llama 3.2) • MongoDB • PostgreSQL • Next.js

- Architected a 4-node LangGraph multi-agent pipeline combining deterministic PostgreSQL filtering with LLM-based ranking; achieved 100% intent extraction accuracy parsing complex user constraints (budget, brand, specs).
- Eliminated search hallucinations by grounding LLM recommendations in real-time verified database context; used Ollama (Llama 3.2) as local fallback ensuring zero downtime on API failures.
- Built a BeautifulSoup scraping pipeline ingesting e-commerce catalog data into PostgreSQL (Supabase) and MongoDB, feeding the recommendation engine with verified product data.

PlantDoc — Crop Disease Detection Web App ↗

2024

FastAPI • PyTorch • Streamlit • ResNet18

- Fine-tuned EfficientNet-B3 on a 54,000-image plant disease dataset; achieved 99% test accuracy across 38 classes with batch upload and real-time inference, enabling farmers to diagnose crop disease from a photo.

EDUCATION

B.S Artificial Intelligence — University of Central Punjab, Lahore

Nov 2022 – Jul 2026

Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Knowledge Representation & Reasoning, Database Systems, Data Structures & Algorithms, Probability & Statistics

FYP: Findify — Multimodal AI Product Discovery Platform (Project Lead)